

BERNE CHU

510-333-3086 | b7chu@ucsd.edu

Summary

I am a second-year Bioinformatics undergraduate at UCSD with experience in biology-related projects, coding, and collaborative teamwork. I am eager to contribute to research, expand my skills, and explore my interests in biology and computational science.

Education

Moreau Catholic High School

2020 – 2024

High School Diploma

University of California, San Diego

2024 – expected graduation: 2027

Bachelors of Science, Bioinformatics | School of Biological Sciences

Relevant coursework: General Biology lab

Current GPA: 3.8

Experiences & Awards

Undergraduate Research Intern — MacRae Lab, Scripps Research

2025–current

- A molecular biology and bioinformatics research lab at Scripps Research.
- I primarily contributed to the dry-lab side of a molecular biology and bioinformatics project. Using R, Pymol, and Excel, I compiled and analyzed AGO2 variants from literature and public databases, performed pathogenicity predictions and VEP integration, and am currently working on mapping the new mutants onto the 3D structure. In the wet lab, prepared buffers and solutions and conducted PCR experiments.

BUMMP Mass Spectrometry Workshop

2025

- A 3-day training at UCSD's Goeddel Family Technology Sandbox, where I gained hands-on experience with mass spectrometry and quantitative proteomics to study protein changes in cyanobacteria under light/dark cycles.

Biology Undergraduate Masters Mentorship Program (BUMMP): Mentee

2024–present

- A program with the goal of providing resources for biology undergraduate students to participate in research and to achieve a successful future in STEM through career and professional development.

Biological Sciences Student Association (BSSA): Mentee

2024–present

- UC San Diego's largest biology organization aimed to help student gain valuable connection and develop professionally.

Scholarships, Inc: Scholarship

2024

- A \$5000 scholarship awarded to me for college by Scholarship, Inc. for outstanding academics.

Science in The Park, County of Alameda: Exhibitor

2022

- A science fair that happens every other year located at CSU East Bay and is open to everyone in the community.
- In October of 2022, I participated in this science fair. My sister and I gave out around 200 silkworm eggs while explaining the life cycle of silkworms and the structure of silk. I did research, made posters, and demonstrated how to extract silk from cocoons. I talked to many visitors from the food industry, the automobile industry, and artists.

Skills

- Through hands-on training from the MS workshop, I learned sample preparation, data acquisition, and analysis techniques using mass spectrometry.
- From the general biology lab, I gain experience in how to use a micropipette, how to conduct PCR, and how to run gel electrophoresis.
- As a Bioinformatics major, through my coursework, I have gained coding experience in Java.
- I am a native speaker of Mandarin and Cantonese, with full fluency in reading and writing in Mandarin.